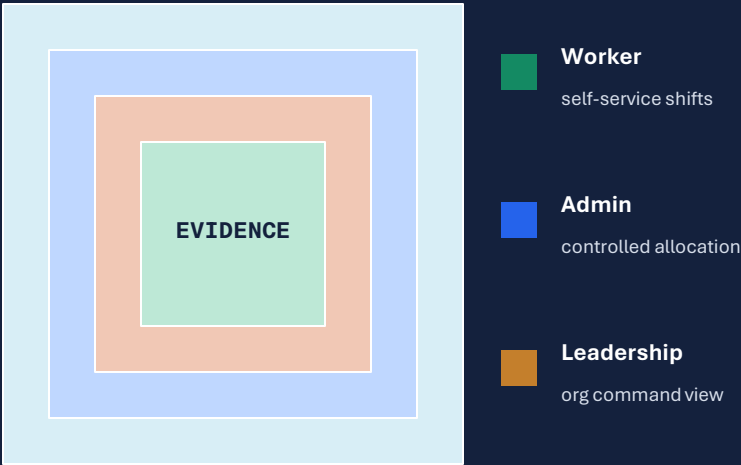


System Architecture Document

Workforce Operating Platform — GlobalSolutions

The platform joins shift scheduling, controlled RDP allocation, live session tracking, quality scoring, leaderboard visibility, payroll evidence, and leadership reporting into one governed operating layer.



9

from scheduling through payroll

30s

machine state refreshed continuously

90d

core control, payroll, quality, command

Manual coordination creates scale risk before the workforce reaches hundreds.

WhatsApp shift tables
Availability is unstructured, hard to approve, and easy to lose.

Manual RDP allocation
No conflict prevention; two workers can be pointed at the same resource.

Disconnected quality data
Training assessments and admin judgment do not compound into worker records.

Spreadsheet payroll
Hours and pay are reconciled after the fact instead of being born from sessions.

SYSTEM RESPONSE

Structured shift approval

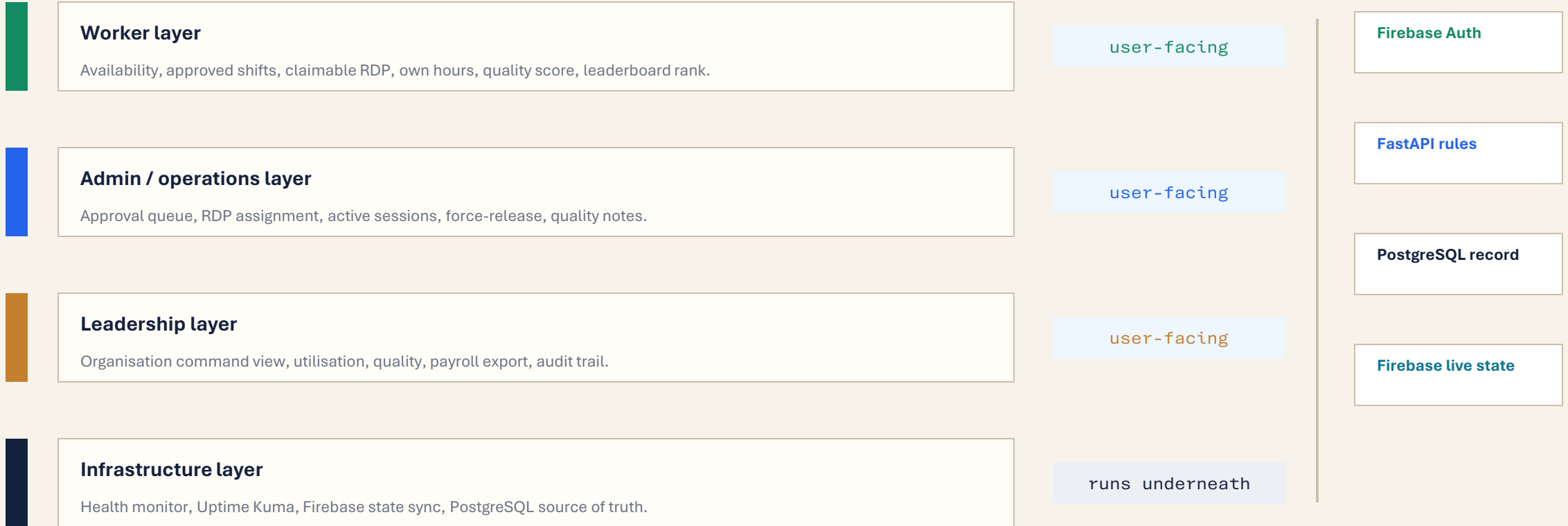
Claim lock and assignment rules

Composite quality score

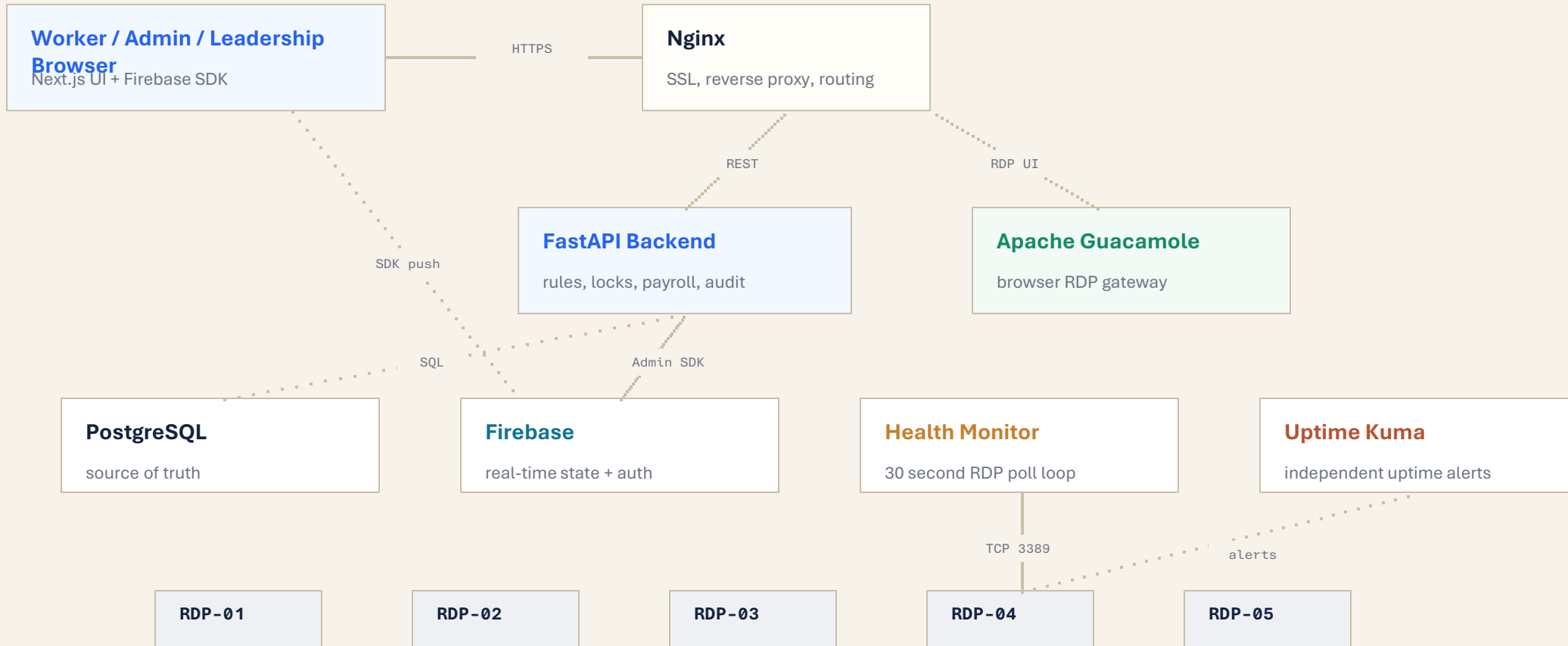
Session-backed payroll bridge

The point is not digitising today's habits. The point is removing the operational fragility that limits scale.

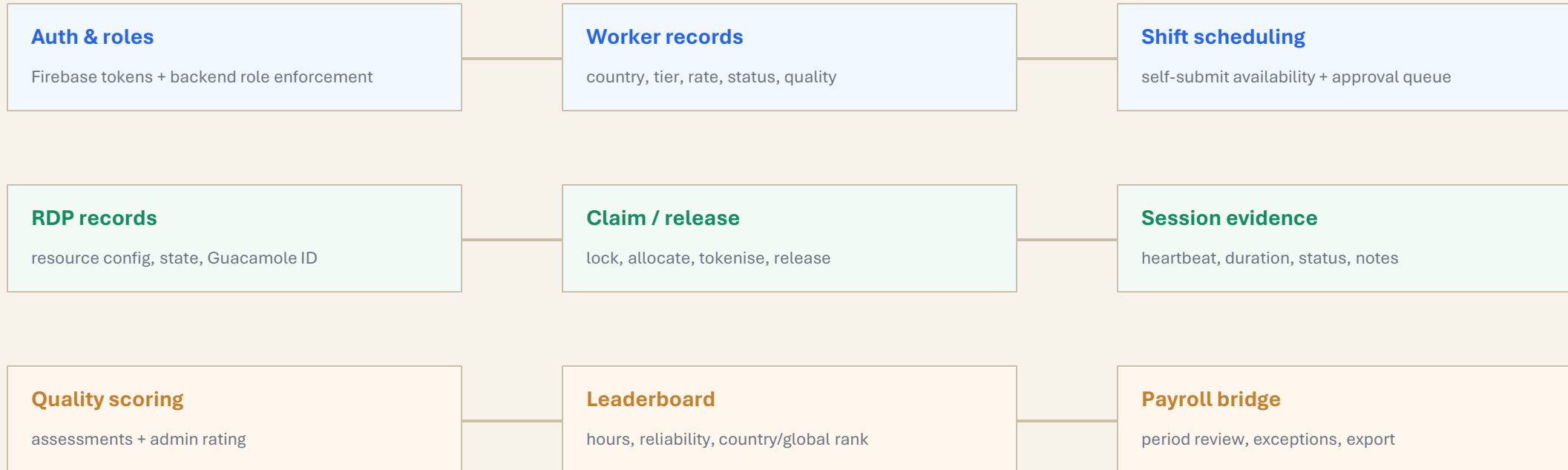
Three user layers share one infrastructure layer, so every role sees the same operating truth.



The architecture separates experience, business rules, real-time state, and durable evidence.

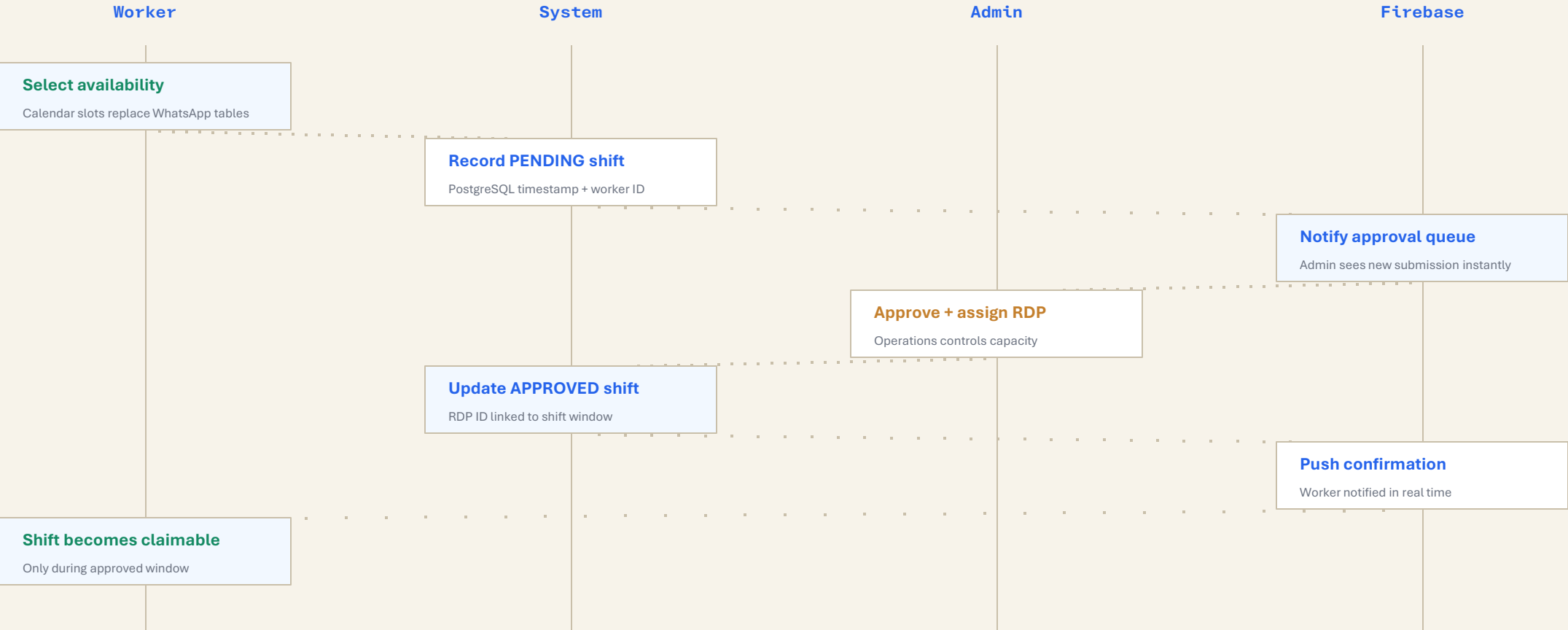


Nine modules cover the worker lifecycle from availability to payment evidence.



Lifecycle rule: every module must either create operational control, improve worker evidence, or feed leadership reporting.

Shift approval turns chat-table intent into controlled, time-bound RDP allocation.



The claim engine prevents double allocation before the session starts.



LOCK GATE

No two workers can hold the same machine because the lock is acquired before the allocation record and state change.

Release path

Worker release, admin force-release, or heartbeat timeout closes Guacamole, writes end time and duration, then returns the RDP to ONLINE_FREE or ASSIGNED.

60s

HEARTBEAT

browser keep-alive cadence

10m

IDLE FLAG

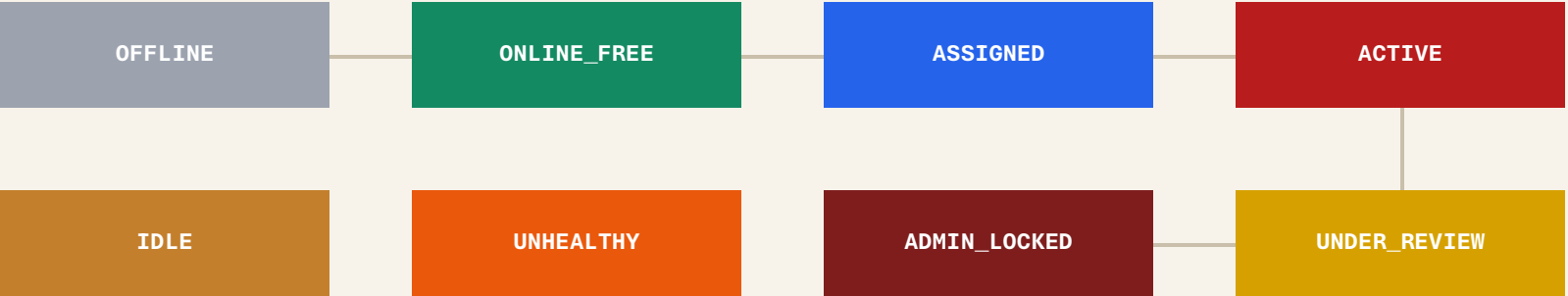
no heartbeat detected

20m

AUTO-RELEASE

resource returned to pool

RDP state is worker-visible, admin-governed, and leadership-auditable.

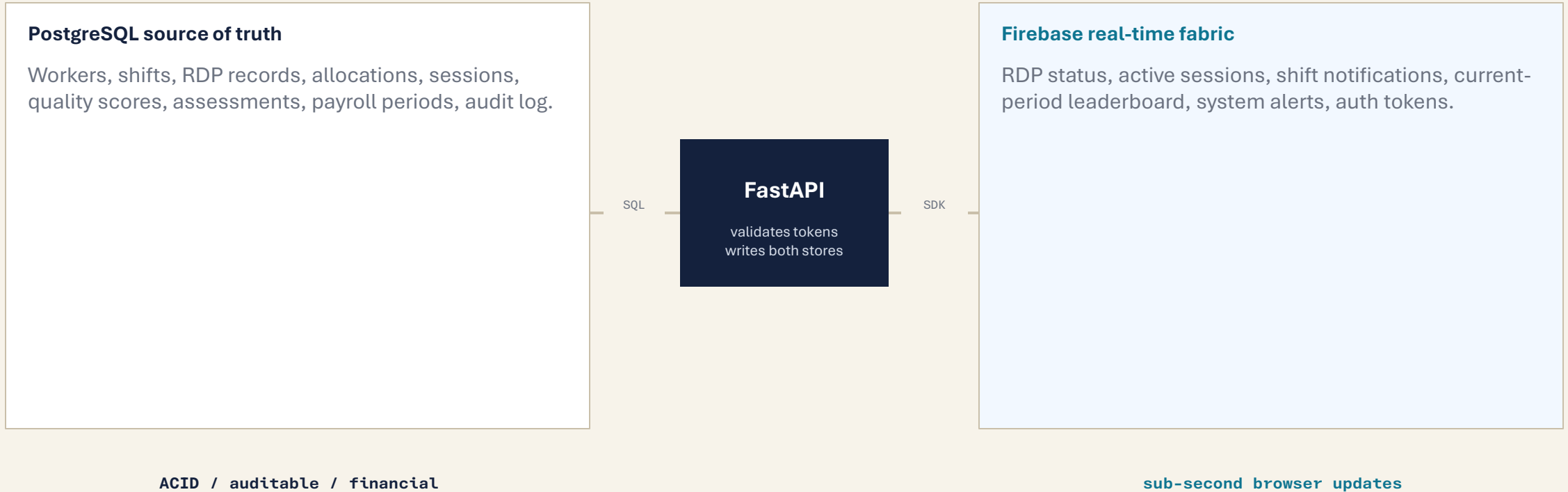


LIVE BOARD SIGNALS

	RDP-01	ACTIVE
	RDP-02	ONLINE_FREE
	RDP-03	ASSIGNED
	RDP-04	IDLE
	RDP-05	OFFLINE

State is written to PostgreSQL and mirrored to Firebase, so the board is fast without losing auditability.

Firestore moves live state; PostgreSQL keeps the company record.

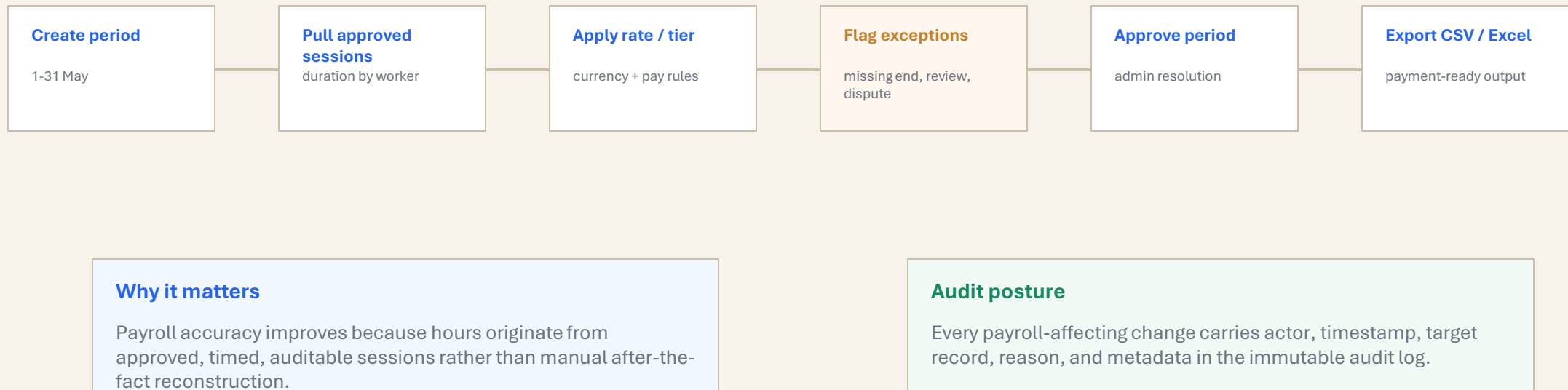


Performance becomes measurable when quality, reliability, and hours share one scorecard.



Gamification is useful only because it rests on verified sessions and accountable ratings.

Payroll becomes an evidence-backed export instead of a manual workbook reconciliation.



The 90-day plan ships control first, intelligence second, and scale third.

